Paola Jeannette Nazate Burgos

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Profile

Researcher and Ph.D. candidate in Electrical Engineering specializing in autonomous robotic systems for unstructured and outdoor environments. Expertise in SLAM and sensor fusion under GNSS-denied conditions, multimodal perception using lidar, radar, RGB, and inertial sensors, and the integration of distributed sensor networks to enhance robotic autonomy. Strong research orientation toward agricultural robotics, environmental monitoring, and smart sensing systems for natural resource management.

Research Expertise

- Agricultural & Field Robotics: Autonomous navigation in orchards and farms; perception-driven mapping for vegetation and phenotypic characterization.
- Robotics & Autonomous Systems: SLAM, sensor fusion, motion control, navigation in unstructured environments and 3D mapping
- Machine Learning & AI: Deep learning, reinforcement learning, supervised and unsupervised learning, probabilistic models.
- Multimodal Perception: lidar, radar, RGB/RGB-D cameras, hyperspectral sensing, IMU, and distributed sensor arrays for environment mapping and semantic interpretation.
- Robotic Systems Implementation: ROS-based development, simulation (Gazebo/RViz), deep learning and reinforcement learning for perception and control.
- Multidisciplinary Applications: Bridging theory and practice in robotics, artificial intelligence, and computational sciences for research and education.

Education

Ph.D. Candidate of Electrical Engineering Expected completion: 2026
Pontificia Universidad Católica de Chile UC *Santiago, Chile*
Dissertation: Localization and sensor fusion for mobile robots in GNSS-denied environments with applications to agricultural robots in arboreal environments.
Additional research activities: Field experimentation in fruit orchards and development of reproducible multimodal datasets (LiDAR, radar, RGB, IMU) avalible on github.com/RAL-UC/Pullally$_{Dataset}$ SLAM and perception research, with open-source code and documentation hosted on github.com/RAL-UC/RoSA$_{SLAM}$.

Master's Degree in University Education 2016 – 2018
Escuela de las Fuerzas Armadas ESPE *Sangolqui, Ecuador*
Thesis: Impact of degree modalities in the dropout rate

Electronic engineer in control and automation 2006 – 2012
Escuela Politécnica Nacional EPN *Quito, Ecuador*
Thesis: Design and construction of a spherical mobile robot with real-time vision

Publications

Agricultural Robotics & Autonomous Systems

- **Nazate-Burgos, P.**, Torres-Torriti, M., Huang, S., Auat Cheein, F. (2026). Consistent lidar-only SLAM for legged agricultural robots in arboreal environments via robust dimensionality reduction. Computers and Electronics in Agriculture, volume 247, ISSN 0168-1699. DOI: 10.1016/j.compag.2026.111687.
- Torres-Torriti, M., **Nazate-Burgos, P**. (2023). SLAM in Agriculture. In: Zhang, Q. (eds) Encyclopedia of Smart Agriculture Technologies. Springer, Cham. DOI: 10.1007/978-3-030-89123-7_235-1.

- Torres-Torriti, M., **Nazate-Burgos, P**. (2023). Field Machinery Automated Guidance. In: Zhang, Q. (eds) Encyclopedia of Smart Agriculture Technologies. Springer, Cham. DOI: 10.1007/978-3-030-89123-7_229-1.
- Torres-Torriti, M., **Nazate-Burgos, P**., Paredes-Lizama, F., Guevara, J. & Auat Cheein F. (2022). Passive Landmark Geometry Optimization and Evaluation for Reliable Autonomous Navigation in Mining Tunnels Using 2D Lidars. Sensors, Vol.22, N°8. DOI: 10.3390/s22083038.

Distributed Sensing & Environmental Monitoring

- Arrais, S., **Nazate-Burgos, P**., Garzón, N. O., Caraguay, Á. L. V., Urquiza-Aguiar, L. (2026). A Lightweight Python Recovery Tool for Waveform Gap Recovery in Seismic–Volcanic Monitoring Networks. Technologies, 14(4), 211. 10.3390/technologies14040211.
- Enríquez López, W., **Nazate-Burgos, P**. (2023). Disposición de Sensores Sísmicos para la Detección de Lahares en el Volcán Cotopaxi Basado en un Datalogger con Field Programmable Gate Arrays. Revista Politécnica, 52(1), 7–14. DOI: 10.33333/rp.vol52n1.01.
- Cárdenas, D., Benítez, D. S., Ruíz, M., Enríquez, W., **Nazate, P**., Ramos, C., Espín, C., & Soria, V. (2021). On the Monitoring of the Electromagnetic Fields Accompanying the Seismic and Volcanic Activity of the Chiles Volcano: Preliminary Results. IEEE Fifth Ecuador Technical Chapters Meeting (ETCM), pp. 1-5, DOI: 10.1109/ETCM53643.2021.9590751.
- Enríquez, W., **Nazate, P**., & Marcillo, O. (2018). Prototipo DAS basado en FPGA de 12 canales para monitoreo geodinámico. Visión Electrónica, 12(1), 73–82. DOI: 10.14483/22484728.13782.

Work Experience

Electronics and instruments researcher Aug 2025 – Present
Instituto Geofísico EPN *Quito, Ecuador*
Duties include: *Web page: www.igepn.edu.ec*

- analyze multi-sensor data to identify patterns related to seismic activity, magma movement, and volcanic gas emissions
- develop automated research reports summarizing volcanic activity trends and seismic risk indicators using open-source tools such as Python and GIS-based visualization software
- design and prototype high-sensitivity seismic array instruments, improving sensor durability under extreme conditions including heat, ash, and vibration

Assistant professor Mar 2020 – Jul 2025
Electrical Deparment UC *Santiago, Chile*
Duties include: *Web page: www.uc.cl*

- assistant teaching of Fundamentals of Robotics
- assistant corrector of Automatic Control
- teaching AI Programming I
- assistant in the accreditation process of the master's degree program 2025
- assistant in the international accreditation process ABET of the bachelor's degree program 2023

Electronics and control engineering Jul 2013 – Jul 2019
Instituto Geofísico EPN *Quito, Ecuador*
Duties include: *Web page: www.igepn.edu.ec*

- data analysis from distributed seismic, infrasonic, and electromagnetic sensor networks for environmental monitoring.
- development of automated data processing pipelines and visualization tools (Python, GIS) for real-time situational awareness.
- design and prototyping of robust sensing hardware for harsh outdoor environments (heat, ash, vibration), relevant to seismic and volcanic systems.
- deployment and maintenance of spatially distributed sensor networks operating as early IoT-like platforms for real-time environmental monitoring and decision support.

Project engineer Jan 2013 – Jun 2013
Unión Eléctrica *Quito, Ecuador*
Duties include: *Web page: www.uniongr.com*

- project management
- specifications of electrical installations
- design and install visual security systems cctv

Languages

Spanish (fluent), English (fluent)

Technical Skills

Programming Languages	:	C/C++, Python, MATLAB, LabVIEW, Arduino.
Mathematical & Simulation Tools	:	Simulink, Proteus, LaTeX, PyGame.
Machine Learning & AI Frameworks	:	TensorFlow, PyTorch, Keras, Scikit-learn, OpenAI Gym.
Computer Vision & Robotics Libraries	:	OpenCV, NumPy, SciPy, Matplotlib, Velodyne-Scan, Unitree-Robotics.
Robotics & Middleware	:	ROS, HELIOS, Gazebo, RViz
Operating Systems	:	Windows, Linux (Ubuntu, ROS-compatible distros)
Development & Version Control	:	Visual Studio Code, Git, GitHub, Docker

Soft Skills

Interdisciplinary Collaboration	:	Ability to integrate knowledge from robotics, computer science, mathematics, and geophysics to solve complex problems and develop innovative solutions.
Problem-Solving	:	Strong capacity to analyze and address challenging research and engineering problems.
Research	:	Experience in research projects, writing proposal funding, coordinating multidisciplinary teams.
Scientific Communication	:	Skilled in writing high-impact journal articles, preparing conference presentations, and effectively conveying complex technical concepts to diverse audiences.
Adaptability	:	Quick to adopt emerging technologies, methodologies, and tools to enhance research output and teaching quality.

Projects

Tree Detection and Mapping for Autonomous Navigation in Fruit Orchards 2024 – 2025
IEEE AESS Radar Challenge 2025 *Krakow, Poland*
Role: Sensors Engineerg - Thesis student *Project number: 11*

* The main goal was to develop a tree detection and mapping system for autonomous navigation in fruit orchards using Radar information and a fusion with camera and imu, by applying beamforming and FMCW signal processing to obtain detailed images enabling the identification and characterization of different types of vegetation.

Robotics challenges for agriculture and mining using mobile manipulators 2019 – 2021
National fund for scientific and technological development *Santiago, Chile*
Role: Thesis student *Project number: 1171760*

* The specific goals can be summarized as: develop online stochastic Bayesian estimators for the inertial parameters and the generalized inertia matrix of skid-steer mobile manipulators whose load can change significantly during the operation in industrial applications (e.g. excavator-type machines and harvesters).

Study of the electromagnetic fields that accompany seismic and volcanic activity 2018 – 2020
Corporación Ecuatoriana para el Desarrollo de la Investigación y la Academia (Cedia) *Quito, Ecuador*
Role: Research

* General Objective: To establish the possible relationship between changes in electromagnetic fields and the premonitory activity of volcanic eruptions, through the study and measurement of existing electromagnetic radiation levels when there is seismic and/or volcanic activity.

Seismic sensor arrays for lahars detection at Cotopaxi volcano 2017 – 2018
Escuela Politécnica Nacional *Quito, Ecuador*
Role: Research

* General Objective: To develop a method to monitor, detect and predict the extent of Lahars in the Cotopaxi volcano by integrating arrays of seismic sensors located in a spatial geometric form to detect seismic signals unnoticed by a single station.

Grants Awarded

* **Internship operating expenses at University of Technology Sydney** ANID
  Santiago, Chile *2023*
* **National doctorate scholarship** Agencia Nacional de Investigación y Desarrollo de Chile ANID
  Santiago, Chile *2021*
* **Development Comprehensive Student Grant** International Center for Theoretical Physics ICTP
  Trieste, Italy *2019*
* **Entrepreneurship for scientists Student Grant** South American Institute for Fundamental Research SAIFR
  São Paulo, Brazil *2016*
* **School on Advanced Instrumentation Student Grant** Instituto Tecnologico de Costa Rica
  Cartago, Costa Rica *2014*

Other Activities

Conference:

* Workshop on Novel Approaches for Precision Agriculture and Forestry with Autonomous Robots. IEEE International Conference on Robotics & Automation (ICRA 2025). ICRA - IEEE. Poster Presentation.
* Workshop on Agricultural Robotics for a Sustainable Future. 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023. IROS - IEEE. Poster Presentation.
* Workshop Present and Future of Agricultural Robotics and Technologies. Part of 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023. IROS - IEEE. Poster Presentation.
* Jornada Técnica AC3E 2023, organized by Universidad Técnica Santa María 2023. AC3E. Poster Presentation.
* UC REsearch Week, organized by Engineering department UC, October 09, 2022. Jornada Técnica. "Reliable autonomous navigation for mobile robots inside mining tunnels". Speaker.
* Women in Science UC, organized by Engineering department UC, 2022. Mujeres UC. Conference attendee
* Technical Day, organized by Centro Avanzado de Ingeniería Eléctrica y Electrónica AC3E, August 19, 2022. Jornada Técnica. Poster Presentation.
* UTS-UC webinar 2022, organized by University of Technology Sydney (UTS) and UC, March 28, 2022. Webinar. Conference attendee and speaker.
* Tartan SLAM Series Fall Edition, organized by Airlab, September 30 to December 09, 2021. https://theairlab.org/tartanslamseries2/. Conference attendee.
* 17th International Conference on Automation Science and Engineering (CASE) 2021, organized by the Institute of Electrical and Electronics Engineers (IEEE), August 23-27, 2021. Conference attendee.

* IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2021, organized by the Institute of Electrical and Electronics Engineers (IEEE), September 27 to October 01, 2021. https://www.iros2021.org/. Conference attendee.
* 7th International Mining Automation Congress Automaning 2020, organized by the Advanced Mining Technology Center of the University of Chile, November 30 and December 4, 2020. https://gecamin.com/automining/. Conference attendee and speaker.

Consulting Projects:

* Study of technologies for worker localization and monitoring. Advanced Center for Electronic and Electrical Engineering. October-December, 2020.

Community Service:

* Reviewer for Journal of Field Robotics (2023).
* Volunteer programming teacher for girls Technovation - Girl (2022).
* Organizing committee Latin American and Caribbean Seismological Commission IV Assembly LACS Ecuador (2022).
* Volunteer IA Ecuador (2021)
* Member of Robotics and Automation Laboratory, Engineering Department, UC